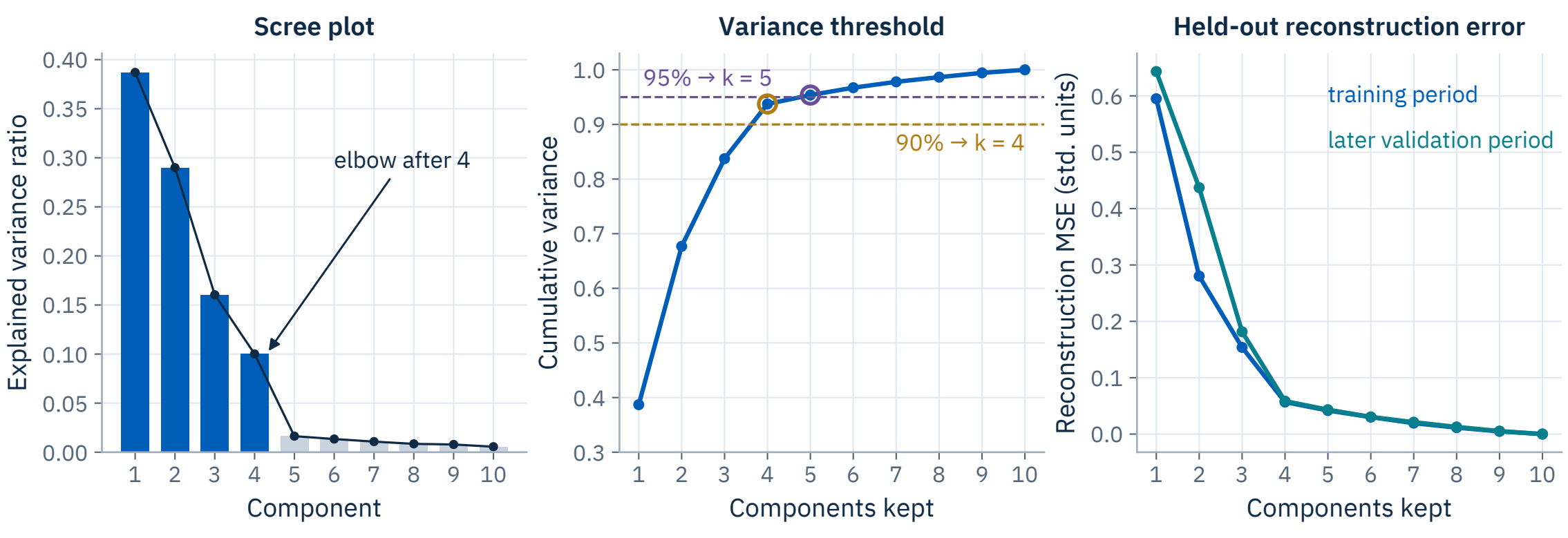
Scree plot
Explained variance ratio
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
elbow after 4
1 2 3 4 5 6 7 8 9 10
Component
Variance threshold
Cumulative variance
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
95% → k = 5
90% → k = 4
1 2 3 4 5 6 7 8 9 10
Components kept
Held-out reconstruction error
Reconstruction MSE (std. units)
0.6
0.5
0.4
0.3
0.2
0.1
0.0
training period
later validation period
1 2 3 4 5 6 7 8 9 10
Components kept